Лабораторна робота №4

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1 Мета

Ознайомитися з методами статистичної обробки даних.

2 Вихідні дані

1. Ознайомтеся з теоретичним матеріалом по даній лабораторній роботі.
2. Сформуйте такі масиви даних вимірювань:

$$A1(30)=[-3N \ldots 8N\,],\quad A2(3,15)=\begin{bmatrix} 1 & 1+h_1 & \ldots & 2N \\ -\frac{N}{2} & -\frac{N}{2}+h_2 & \ldots & \frac{N}{2} \\ N-10 & N-10+h_3 & \ldots & N+10 \end{bmatrix},$$

де N – номер варіанту, $h_i=\dfrac{\left(A2_{i15}-A2_{i1}\right)}{14}$.

3. Виконайте статичний аналіз та обробку цих даних засобами системи *Matlab*.
4. Звіт по виконаній лабораторній роботі здати викладачу.

3 Отримані результати

Скрипт

```
clc
format shortg
N=input('Vvedit N = ');
A1=linspace(-3*N,8*N,30);
h1=((2*N)-1)/14;
h2=((N/2)-(-N/2))/14;
h3=((N+10)-(N-10))/14;
A2=[1:h1:2*N; -N/2:h2:N/2; N-10:h3:N+10];
B1=10*randn(1,30);
B2=10*randn(3,15);
k=0;
start=true;
```

```
while start
  k=k+1;
  if k > 1
    ans_user = input('Bazhaete prodovzhiti?(Y/N) ', 's');
      if ~strcmpi(ans_user, 'Y')
        disp('Prohrama zavershena.');
        break;
      end
  end
  punkt=input('Vvedit bazhanyi punkt(1, 2, 3.3, 4) ');
switch punkt
  case 1
  clc
  printf('\n////////MATRIX////////\n\n')
  A1
  A2
  B1
  B2
  printf('\n////////Max////////\n\n')
  A1max=max(A1)
  A1B1max=max(A1,B1)
  [H1max,I1max]=max(A1)
  A2max=max(A2)
  A2B2max=max(A2,B2)
  [H2max,I2max]=max(A2)
  A2max1=max(A2,[],1)
  A2max2=max(A2,[],2)
  printf('\n////////Min////////\n\n')
  A1min=min(A1)
  A1B1min=min(A1,B1)
  [H1min,I1min]=min(A1)
  A2min=min(A2)
  A2B2min=min(A2,B2)
  [H2min,I2min]=min(A2)
  A2min1=min(A2,[],1)
  A2min2=min(A2,[],2)

  case 2
  clc
  printf('\n////////MATRIX////////\n\n')
  A1
  A2
  printf('\n/////////MEAN/////////\n\n')
  A1mn=mean(A1)
  A2mn1=mean(A2)
  A2mn2=mean(A2,1)
  A2mn3=mean(A2,2)
  printf('\n////////MEDIAN////////\n\n')
  A1md=median(A1)
  A2md1=median(A2)
  A2md2=median(A2,1)
  A2md3=median(A2,2)
  printf('\n//////////STD/////////\n\n')
  A1std1=std(A1)
```

```
  A1std2=std(A1,0)
  A1std3=std(A1,1)
  A2std1=std(A2)
  A2std2=std(A2,0)
  A2std3=std(A2,1)
  A2std4=std(A2,0,1)
  A2std5=std(A2,0,2)
  A2std6=std(A2,1,1)
  A2std7=std(A2,1,2)

  case {3, 3.3}
  clc
  printf('\n////////MATRIX////////\n\n')
  A11=[A1+B1*i]
  A12=[A1-B1*i]
  A3=[A1;A11;A12]
  printf('\n///////CPLXPAIR///////\n\n')
  A1112cp=cplxpair([A11,A12])
  A3cp1=cplxpair(A3)
  A3cp2=cplxpair(A3,[],1)
  A3cp3=cplxpair(A3',[],2)

  case 4
  clc
  printf('\n////////MATRIX////////\n\n')
  A1
  A2
  printf('\n//////////SUM////////\n\n')
  A1sum=sum(A1)
  A2sum1=sum(A2)
  A2sum2=sum(A2,1)
  A2sum3=sum(A2,2)
  printf('\n//////////PROD////////\n\n')
  A1prd=prod(A1)
  A2prd1=prod(A2)
  A2prd2=prod(A2,1)
  A2prd3=prod(A2,2)
  printf('\n////////CUMSUM////////\n\n')
  A1csum=cumsum(A1)
  A2csum1=cumsum(A2)
  A2csum2=cumsum(A2,1)
  A2csum3=cumsum(A2,2)
  printf('\n///////CUMPROD////////\n\n')
  A1cprd=cumprod(A1)
  A2cprd1=cumprod(A2)
  A2cprd2=cumprod(A2,1)
  A2cprd3=cumprod(A2,2)
  printf('\n//////////DIFF////////\n\n')
  A1df=diff(A1)
  A2df=diff(A2)

otherwise
  disp('Nevirnyi punkt! Vyberit 1, 2, 3.3 abo 4.');
```

```
  end
end
```

Результат 1-го "Кейсу"

```
Vvedit N = 10
Vvedit bazhanyi punkt(1, 2, 3.3, 4) 1

/////////MATRIX////////

A1 =

 Columns 1 through 7:

          -30      -26.207      -22.414      -18.621      -14.828      -11.034      -7.2414

 Columns 8 through 14:

      -3.4483      0.34483       4.1379        7.931       11.724       15.517        19.31

 Columns 15 through 21:

       23.103       26.897        30.69       34.483       38.276       42.069       45.862

 Columns 22 through 28:

       49.655       53.448       57.241       61.034       64.828       68.621       72.414

 Columns 29 and 30:

       76.207           80

A2 =

 Columns 1 through 7:

            1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
           -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429
            0       1.4286       2.8571       4.2857       5.7143       7.1429       8.5714

 Columns 8 through 14:

         10.5       11.857       13.214       14.571       15.929       17.286       18.643
            0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857
           10       11.429       12.857       14.286       15.714       17.143       18.571

 Column 15:

           20
            5
           20
```

```
B1 =

 Columns 1 through 7:

      10.644       11.847       -8.422       7.5787      -5.4151       9.3489      -7.2253

 Columns 8 through 14:

     -9.1608      -4.4224       2.1687        17.56      -2.7867      -7.8543         17.6

 Columns 15 through 21:

      16.439      0.26701       5.9705      -0.9121      -7.4622       11.654       9.5875

 Columns 22 through 28:

      7.1177      -13.342       -11.12      -4.5304       1.0967      -9.4345        14.08

 Columns 29 and 30:

       7.238     0.076706

B2 =

 Columns 1 through 7:

      9.2081      -6.8587      -14.732       9.1546      -3.1897       6.1515       3.0829
      3.7808        4.524      -24.034      -2.6484      -12.333       13.074      -10.548
      -11.95       12.974      -1.7788      -9.0815      -4.2309       12.286       7.4919

 Columns 8 through 14:

    -0.26838       13.297     -0.58767      -25.202       7.5814      -6.5271      -15.368
     -11.545      -19.306      -5.4793       4.3528       8.0323       12.337       1.6801
       -11.2      -17.856      -3.5855     -0.95116      -3.3218       15.006       -4.028

 Column 15:

     -4.3685
     -5.6954
     0.39055


/////////Max/////////

A1max =    80
A1B1max =

 Columns 1 through 7:

      10.644       11.847       -8.422       7.5787      -5.4151       9.3489      -7.2253

 Columns 8 through 14:
```

```
      -3.4483      0.34483       4.1379        17.56       11.724       15.517        19.31

 Columns 15 through 21:

       23.103       26.897        30.69       34.483       38.276       42.069       45.862

 Columns 22 through 28:

       49.655       53.448       57.241       61.034       64.828       68.621       72.414

 Columns 29 and 30:

       76.207           80

H1max =    80
I1max =    30
A2max =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429

 Columns 8 through 14:

         10.5       11.857       13.214       14.571       15.929       17.286       18.643

 Column 15:

           20

A2B2max =

 Columns 1 through 7:

       9.2081       2.3571       3.7143       9.1546       6.4286       7.7857       9.1429
       3.7808        4.524      -3.5714      -2.6484      -2.1429       13.074     -0.71429
            0       12.974       2.8571       4.2857       5.7143       12.286       8.5714

 Columns 8 through 14:

         10.5       13.297       13.214       14.571       15.929       17.286       18.643
            0      0.71429       1.4286       4.3528       8.0323       12.337       4.2857
           10       11.429       12.857       14.286       15.714       17.143       18.571

 Column 15:

           20
            5
           20

H2max =

 Columns 1 through 7:
```

```
            1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643

 Column 15:

           20

I2max =

 Columns 1 through 7:

            1            1            1            1            1            1            1

 Columns 8 through 14:

            1            1            1            1            1            1            1

 Column 15:

            1

A2max1 =

 Columns 1 through 7:

            1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643

 Column 15:

           20

A2max2 =

           20
            5
           20


/////////Min////////

A1min =   -30
A1B1min =

 Columns 1 through 7:

         -30      -26.207      -22.414      -18.621      -14.828      -11.034      -7.2414
```

```
 Columns 8 through 14:

      -9.1608      -4.4224       2.1687        7.931      -2.7867      -7.8543         17.6

 Columns 15 through 21:

       16.439      0.26701       5.9705      -0.9121      -7.4622       11.654       9.5875

 Columns 22 through 28:

       7.1177      -13.342       -11.12      -4.5304       1.0967      -9.4345        14.08

 Columns 29 and 30:

        7.238     0.076706

H1min =    -30
I1min =      1
A2min =

 Columns 1 through 7:

           -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429

 Columns 8 through 14:

            0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857

 Column 15:

            5

A2B2min =

 Columns 1 through 7:

            1      -6.8587      -14.732       5.0714      -3.1897       6.1515       3.0829
           -5      -4.2857      -24.034      -2.8571      -12.333      -1.4286      -10.548
       -11.95       1.4286      -1.7788      -9.0815      -4.2309       7.1429       7.4919

 Columns 8 through 14:

     -0.26838       11.857     -0.58767      -25.202       7.5814      -6.5271      -15.368
      -11.545      -19.306      -5.4793       2.1429       2.8571       3.5714       1.6801
        -11.2      -17.856      -3.5855     -0.95116      -3.3218       15.006      -4.028

 Column 15:

      -4.3685
      -5.6954
      0.39055

H2min =
```

```
 Columns 1 through 7:

          -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429

 Columns 8 through 14:

           0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857

 Column 15:

           5

I2min =

 Columns 1 through 7:

           2            2            2            2            2            2            2

 Columns 8 through 14:

           2            2            2            2            2            2            2

 Column 15:

           2

A2min1 =

 Columns 1 through 7:

          -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429

 Columns 8 through 14:

           0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857

 Column 15:

           5

A2min2 =

           1
          -5
           0
```

Результат 2-го "Кейсу"

```
Bazhaete prodovzhiti?(Y/N) y
Vvedit bazhanyi punkt(1, 2, 3.3, 4) 2

/////////MATRIX////////
```

```
A1 =

 Columns 1 through 7:

         -30      -26.207      -22.414      -18.621      -14.828      -11.034      -7.2414

 Columns 8 through 14:

      -3.4483      0.34483       4.1379        7.931       11.724       15.517        19.31

 Columns 15 through 21:

       23.103       26.897        30.69       34.483       38.276       42.069       45.862

 Columns 22 through 28:

       49.655       53.448       57.241       61.034       64.828       68.621       72.414

 Columns 29 and 30:

       76.207           80

A2 =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429
           0       1.4286       2.8571       4.2857       5.7143       7.1429       8.5714

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
           0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857
          10       11.429       12.857       14.286       15.714       17.143       18.571

 Column 15:

          20
           5
          20


//////////MEAN//////////

A1mn =    25
A2mn1 =

 Columns 1 through 7:

      -1.3333     -0.16667            1       2.1667       3.3333          4.5       5.6667

 Columns 8 through 14:
```

```
      6.8333            8       9.1667       10.333         11.5       12.667       13.833

 Column 15:

          15

A2mn2 =

 Columns 1 through 7:

     -1.3333     -0.16667            1       2.1667       3.3333          4.5       5.6667

 Columns 8 through 14:

      6.8333            8       9.1667       10.333         11.5       12.667       13.833

 Column 15:

          15

A2mn3 =

        10.5
   1.4803e-16
          10


/////////MEDIAN////////

A1md =    25
A2md1 =

 Columns 1 through 7:

           0       1.4286       2.8571       4.2857       5.7143       7.1429       8.5714

 Columns 8 through 14:

          10       11.429       12.857       14.286       15.714       17.143       18.571

 Column 15:

          20

A2md2 =

 Columns 1 through 7:

           0       1.4286       2.8571       4.2857       5.7143       7.1429       8.5714

 Columns 8 through 14:

          10       11.429       12.857       14.286       15.714       17.143       18.571
```

```
 Column 15:

          20

A2md3 =

         10.5
            0
           10


//////////STD/////////

A1std1 = 33.392
A1std2 = 33.392
A1std3 = 32.831
A2std1 =

 Columns 1 through 7:

      3.2146       3.5973       3.9821       4.3684       4.7559       5.1443       5.5334

 Columns 8 through 14:

      5.9231       6.3133       6.7038       7.0946       7.4857        7.877       8.2686

 Column 15:

      8.6603

A2std2 =

 Columns 1 through 7:

      3.2146       3.5973       3.9821       4.3684       4.7559       5.1443       5.5334

 Columns 8 through 14:

      5.9231       6.3133       6.7038       7.0946       7.4857        7.877       8.2686

 Column 15:

      8.6603

A2std3 =

 Columns 1 through 7:

      2.6247       2.9372       3.2514       3.5668       3.8832       4.2003        4.518

 Columns 8 through 14:

      4.8362       5.1547       5.4736       5.7927        6.112       6.4316       6.7512
```

```
 Column 15:

       7.0711

A2std4 =

 Columns 1 through 7:

       3.2146       3.5973       3.9821       4.3684       4.7559       5.1443       5.5334

 Columns 8 through 14:

       5.9231       6.3133       6.7038       7.0946       7.4857        7.877       8.2686

 Column 15:

       8.6603

A2std5 =

       6.0693
       3.1944
       6.3888

A2std6 =

 Columns 1 through 7:

       2.6247       2.9372       3.2514       3.5668       3.8832       4.2003        4.518

 Columns 8 through 14:

       4.8362       5.1547       5.4736       5.7927        6.112       6.4316       6.7512

 Column 15:

       7.0711

A2std7 =

       5.8635
       3.0861
       6.1721
```

Результат 3-го "Кейсу"

```
Bazhaete prodovzhiti?(Y/N) y
Vvedit bazhanyi punkt(1, 2, 3.3, 4) 3

/////////MATRIX/////////
```

```
A11 =

 Columns 1 through 3:

         -30 +      10.644i      -26.207 +      11.847i      -22.414 -       8.422i

 Columns 4 through 6:

      -18.621 +      7.5787i      -14.828 -      5.4151i      -11.034 +      9.3489i

 Columns 7 through 9:

      -7.2414 -      7.2253i      -3.4483 -      9.1608i      0.34483 -      4.4224i

 Columns 10 through 12:

       4.1379 +      2.1687i        7.931 +       17.56i       11.724 -      2.7867i

 Columns 13 through 15:

       15.517 -      7.8543i        19.31 +        17.6i       23.103 +      16.439i

 Columns 16 through 18:

       26.897 +     0.26701i        30.69 +      5.9705i       34.483 -      0.9121i

 Columns 19 through 21:

       38.276 -      7.4622i       42.069 +      11.654i       45.862 +      9.5875i

 Columns 22 through 24:

       49.655 +      7.1177i       53.448 -      13.342i       57.241 -       11.12i

 Columns 25 through 27:

       61.034 -      4.5304i       64.828 +      1.0967i       68.621 -      9.4345i

 Columns 28 through 30:

       72.414 +       14.08i       76.207 +       7.238i           80 +    0.076706i

A12 =

 Columns 1 through 3:

         -30 -      10.644i      -26.207 -      11.847i      -22.414 +       8.422i

 Columns 4 through 6:

      -18.621 -      7.5787i      -14.828 +      5.4151i      -11.034 -      9.3489i

 Columns 7 through 9:
```

```
      -7.2414 +      7.2253i      -3.4483 +      9.1608i      0.34483 +      4.4224i

  Columns 10 through 12:

       4.1379 -      2.1687i        7.931 -       17.56i       11.724 +      2.7867i

  Columns 13 through 15:

       15.517 +      7.8543i        19.31 -        17.6i       23.103 -      16.439i

  Columns 16 through 18:

       26.897 -     0.26701i        30.69 -      5.9705i       34.483 +      0.9121i

  Columns 19 through 21:

       38.276 +      7.4622i       42.069 -      11.654i       45.862 -      9.5875i

  Columns 22 through 24:

       49.655 -      7.1177i       53.448 +      13.342i       57.241 +       11.12i

  Columns 25 through 27:

       61.034 +      4.5304i       64.828 -      1.0967i       68.621 +      9.4345i

  Columns 28 through 30:

       72.414 -       14.08i       76.207 -       7.238i           80 -    0.076706i

A3 =

  Columns 1 through 3:

          -30 +           0i      -26.207 +           0i      -22.414 +           0i
          -30 +      10.644i      -26.207 +      11.847i      -22.414 -       8.422i
          -30 -      10.644i      -26.207 -      11.847i      -22.414 +       8.422i

  Columns 4 through 6:

      -18.621 +           0i      -14.828 +           0i      -11.034 +           0i
      -18.621 +      7.5787i      -14.828 -      5.4151i      -11.034 +      9.3489i
      -18.621 -      7.5787i      -14.828 +      5.4151i      -11.034 -      9.3489i

  Columns 7 through 9:

      -7.2414 +           0i      -3.4483 +           0i      0.34483 +           0i
      -7.2414 -      7.2253i      -3.4483 -      9.1608i      0.34483 -      4.4224i
      -7.2414 +      7.2253i      -3.4483 +      9.1608i      0.34483 +      4.4224i

  Columns 10 through 12:

       4.1379 +           0i        7.931 +           0i       11.724 +           0i
       4.1379 +      2.1687i        7.931 +       17.56i       11.724 -      2.7867i
```

```
      4.1379 -      2.1687i        7.931 -       17.56i       11.724 +      2.7867i

 Columns 13 through 15:

      15.517 +           0i       19.31 +           0i       23.103 +           0i
      15.517 -      7.8543i       19.31 +        17.6i       23.103 +      16.439i
      15.517 +      7.8543i       19.31 -        17.6i       23.103 -      16.439i

 Columns 16 through 18:

      26.897 +           0i       30.69 +           0i       34.483 +           0i
      26.897 +     0.26701i       30.69 +      5.9705i       34.483 -      0.9121i
      26.897 -     0.26701i       30.69 -      5.9705i       34.483 +      0.9121i

 Columns 19 through 21:

      38.276 +           0i       42.069 +           0i       45.862 +           0i
      38.276 -      7.4622i       42.069 +      11.654i       45.862 +      9.5875i
      38.276 +      7.4622i       42.069 -      11.654i       45.862 -      9.5875i

 Columns 22 through 24:

      49.655 +           0i       53.448 +           0i       57.241 +           0i
      49.655 +      7.1177i       53.448 -      13.342i       57.241 -       11.12i
      49.655 -      7.1177i       53.448 +      13.342i       57.241 +       11.12i

 Columns 25 through 27:

      61.034 +           0i       64.828 +           0i       68.621 +           0i
      61.034 -      4.5304i       64.828 +      1.0967i       68.621 -      9.4345i
      61.034 +      4.5304i       64.828 -      1.0967i       68.621 +      9.4345i

 Columns 28 through 30:

      72.414 +           0i       76.207 +           0i           80 +           0i
      72.414 +       14.08i       76.207 +       7.238i           80 +    0.076706i
      72.414 -       14.08i       76.207 -       7.238i           80 -    0.076706i


////////CPLXPAIR////////

A1112cp =

 Columns 1 through 3:

         -30 -      10.644i          -30 +      10.644i      -26.207 -      11.847i

 Columns 4 through 6:

     -26.207 +      11.847i      -22.414 -       8.422i      -22.414 +       8.422i

 Columns 7 through 9:

     -18.621 -      7.5787i      -18.621 +      7.5787i      -14.828 -      5.4151i
```

```
Columns 10 through 12:

     -14.828 +      5.4151i      -11.034 -      9.3489i      -11.034 +      9.3489i

Columns 13 through 15:

     -7.2414 -      7.2253i      -7.2414 +      7.2253i      -3.4483 -      9.1608i

Columns 16 through 18:

     -3.4483 +      9.1608i      0.34483 -      4.4224i      0.34483 +      4.4224i

Columns 19 through 21:

      4.1379 -      2.1687i       4.1379 +      2.1687i        7.931 -       17.56i

Columns 22 through 24:

       7.931 +       17.56i       11.724 -      2.7867i       11.724 +      2.7867i

Columns 25 through 27:

      15.517 -      7.8543i       15.517 +      7.8543i        19.31 -        17.6i

Columns 28 through 30:

       19.31 +        17.6i       23.103 -      16.439i       23.103 +      16.439i

Columns 31 through 33:

      26.897 -     0.26701i       26.897 +     0.26701i        30.69 -      5.9705i

Columns 34 through 36:

       30.69 +      5.9705i       34.483 -      0.9121i       34.483 +      0.9121i

Columns 37 through 39:

      38.276 -      7.4622i       38.276 +      7.4622i       42.069 -      11.654i

Columns 40 through 42:

      42.069 +      11.654i       45.862 -      9.5875i       45.862 +      9.5875i

Columns 43 through 45:

      49.655 -      7.1177i       49.655 +      7.1177i       53.448 -      13.342i

Columns 46 through 48:

      53.448 +      13.342i       57.241 -       11.12i       57.241 +       11.12i

Columns 49 through 51:
```

```
      61.034 -      4.5304i       61.034 +      4.5304i       64.828 -      1.0967i

  Columns 52 through 54:

      64.828 +      1.0967i       68.621 -      9.4345i       68.621 +      9.4345i

  Columns 55 through 57:

      72.414 -       14.08i       72.414 +       14.08i       76.207 -       7.238i

  Columns 58 through 60:

      76.207 +       7.238i           80 -    0.076706i           80 +    0.076706i

A3cp1 =

  Columns 1 through 3:

         -30 -      10.644i      -26.207 -      11.847i      -22.414 -       8.422i
         -30 +      10.644i      -26.207 +      11.847i      -22.414 +       8.422i
         -30 +           0i      -26.207 +           0i      -22.414 +           0i

  Columns 4 through 6:

     -18.621 -      7.5787i      -14.828 -      5.4151i      -11.034 -      9.3489i
     -18.621 +      7.5787i      -14.828 +      5.4151i      -11.034 +      9.3489i
     -18.621 +           0i      -14.828 +           0i      -11.034 +           0i

  Columns 7 through 9:

     -7.2414 -      7.2253i      -3.4483 -      9.1608i      0.34483 -      4.4224i
     -7.2414 +      7.2253i      -3.4483 +      9.1608i      0.34483 +      4.4224i
     -7.2414 +           0i      -3.4483 +           0i      0.34483 +           0i

  Columns 10 through 12:

      4.1379 -      2.1687i        7.931 -       17.56i       11.724 -      2.7867i
      4.1379 +      2.1687i        7.931 +       17.56i       11.724 +      2.7867i
      4.1379 +           0i        7.931 +           0i       11.724 +           0i

  Columns 13 through 15:

      15.517 -      7.8543i        19.31 -        17.6i       23.103 -      16.439i
      15.517 +      7.8543i        19.31 +        17.6i       23.103 +      16.439i
      15.517 +           0i        19.31 +           0i       23.103 +           0i

  Columns 16 through 18:

      26.897 -     0.26701i        30.69 -      5.9705i       34.483 -      0.9121i
      26.897 +     0.26701i        30.69 +      5.9705i       34.483 +      0.9121i
      26.897 +           0i        30.69 +           0i       34.483 +           0i

  Columns 19 through 21:
```

```
      38.276 -      7.4622i       42.069 -      11.654i       45.862 -      9.5875i
      38.276 +      7.4622i       42.069 +      11.654i       45.862 +      9.5875i
      38.276 +           0i       42.069 +           0i       45.862 +           0i

  Columns 22 through 24:

      49.655 -      7.1177i       53.448 -      13.342i       57.241 -       11.12i
      49.655 +      7.1177i       53.448 +      13.342i       57.241 +       11.12i
      49.655 +           0i       53.448 +           0i       57.241 +           0i

  Columns 25 through 27:

      61.034 -      4.5304i       64.828 -      1.0967i       68.621 -      9.4345i
      61.034 +      4.5304i       64.828 +      1.0967i       68.621 +      9.4345i
      61.034 +           0i       64.828 +           0i       68.621 +           0i

  Columns 28 through 30:

      72.414 -       14.08i       76.207 -       7.238i           80 -    0.076706i
      72.414 +       14.08i       76.207 +       7.238i           80 +    0.076706i
      72.414 +           0i       76.207 +           0i           80 +           0i

A3cp2 =

  Columns 1 through 3:

         -30 -      10.644i      -26.207 -      11.847i      -22.414 -       8.422i
         -30 +      10.644i      -26.207 +      11.847i      -22.414 +       8.422i
         -30 +           0i      -26.207 +           0i      -22.414 +           0i

  Columns 4 through 6:

     -18.621 -      7.5787i      -14.828 -      5.4151i      -11.034 -      9.3489i
     -18.621 +      7.5787i      -14.828 +      5.4151i      -11.034 +      9.3489i
     -18.621 +           0i      -14.828 +           0i      -11.034 +           0i

  Columns 7 through 9:

     -7.2414 -      7.2253i      -3.4483 -      9.1608i      0.34483 -      4.4224i
     -7.2414 +      7.2253i      -3.4483 +      9.1608i      0.34483 +      4.4224i
     -7.2414 +           0i      -3.4483 +           0i      0.34483 +           0i

  Columns 10 through 12:

      4.1379 -      2.1687i        7.931 -       17.56i       11.724 -      2.7867i
      4.1379 +      2.1687i        7.931 +       17.56i       11.724 +      2.7867i
      4.1379 +           0i        7.931 +           0i       11.724 +           0i

  Columns 13 through 15:

      15.517 -      7.8543i        19.31 -        17.6i       23.103 -      16.439i
      15.517 +      7.8543i        19.31 +        17.6i       23.103 +      16.439i
      15.517 +           0i        19.31 +           0i       23.103 +           0i
```

```
  Columns 16 through 18:

      26.897 -     0.26701i       30.69 -      5.9705i       34.483 -      0.9121i
      26.897 +     0.26701i       30.69 +      5.9705i       34.483 +      0.9121i
      26.897 +           0i       30.69 +           0i       34.483 +           0i

  Columns 19 through 21:

      38.276 -      7.4622i       42.069 -      11.654i       45.862 -      9.5875i
      38.276 +      7.4622i       42.069 +      11.654i       45.862 +      9.5875i
      38.276 +           0i       42.069 +           0i       45.862 +           0i

  Columns 22 through 24:

      49.655 -      7.1177i       53.448 -      13.342i       57.241 -       11.12i
      49.655 +      7.1177i       53.448 +      13.342i       57.241 +       11.12i
      49.655 +           0i       53.448 +           0i       57.241 +           0i

  Columns 25 through 27:

      61.034 -      4.5304i       64.828 -      1.0967i       68.621 -      9.4345i
      61.034 +      4.5304i       64.828 +      1.0967i       68.621 +      9.4345i
      61.034 +           0i       64.828 +           0i       68.621 +           0i

  Columns 28 through 30:

      72.414 -       14.08i       76.207 -       7.238i           80 -    0.076706i
      72.414 +       14.08i       76.207 +       7.238i           80 +    0.076706i
      72.414 +           0i       76.207 +           0i           80 +           0i

A3cp3 =

         -30 -      10.644i          -30 +      10.644i          -30 +           0i
      -26.207 -      11.847i       -26.207 +      11.847i       -26.207 +           0i
      -22.414 -       8.422i       -22.414 +       8.422i       -22.414 +           0i
      -18.621 -      7.5787i       -18.621 +      7.5787i       -18.621 +           0i
      -14.828 -      5.4151i       -14.828 +      5.4151i       -14.828 +           0i
      -11.034 -      9.3489i       -11.034 +      9.3489i       -11.034 +           0i
      -7.2414 -      7.2253i       -7.2414 +      7.2253i       -7.2414 +           0i
      -3.4483 -      9.1608i       -3.4483 +      9.1608i       -3.4483 +           0i
     0.34483 -      4.4224i      0.34483 +      4.4224i      0.34483 +           0i
      4.1379 -      2.1687i       4.1379 +      2.1687i       4.1379 +           0i
       7.931 -       17.56i        7.931 +       17.56i        7.931 +           0i
      11.724 -      2.7867i       11.724 +      2.7867i       11.724 +           0i
      15.517 -      7.8543i       15.517 +      7.8543i       15.517 +           0i
       19.31 -        17.6i        19.31 +        17.6i        19.31 +           0i
      23.103 -      16.439i       23.103 +      16.439i       23.103 +           0i
      26.897 -     0.26701i       26.897 +     0.26701i       26.897 +           0i
       30.69 -      5.9705i        30.69 +      5.9705i        30.69 +           0i
      34.483 -      0.9121i       34.483 +      0.9121i       34.483 +           0i
      38.276 -      7.4622i       38.276 +      7.4622i       38.276 +           0i
      42.069 -      11.654i       42.069 +      11.654i       42.069 +           0i
      45.862 -      9.5875i       45.862 +      9.5875i       45.862 +           0i
```

```
      49.655 -      7.1177i       49.655 +      7.1177i       49.655 +           0i
      53.448 -      13.342i       53.448 +      13.342i       53.448 +           0i
      57.241 -       11.12i       57.241 +       11.12i       57.241 +           0i
      61.034 -      4.5304i       61.034 +      4.5304i       61.034 +           0i
      64.828 -      1.0967i       64.828 +      1.0967i       64.828 +           0i
      68.621 -      9.4345i       68.621 +      9.4345i       68.621 +           0i
      72.414 -       14.08i       72.414 +       14.08i       72.414 +           0i
      76.207 -       7.238i       76.207 +       7.238i       76.207 +           0i
          80 -    0.076706i           80 +    0.076706i           80 +           0i
```

Результат 4-го "Кейсу"

```
Bazhaete prodovzhiti?(Y/N) y
Vvedit bazhanyi punkt(1, 2, 3.3, 4) 4

/////////MATRIX////////

A1 =

 Columns 1 through 7:

         -30      -26.207      -22.414      -18.621      -14.828      -11.034      -7.2414

 Columns 8 through 14:

      -3.4483      0.34483       4.1379        7.931       11.724       15.517        19.31

 Columns 15 through 21:

       23.103       26.897        30.69       34.483       38.276       42.069       45.862

 Columns 22 through 28:

       49.655       53.448       57.241       61.034       64.828       68.621       72.414

 Columns 29 and 30:

       76.207           80

A2 =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -5      -4.2857      -3.5714      -2.8571      -2.1429      -1.4286     -0.71429
           0       1.4286       2.8571       4.2857       5.7143       7.1429       8.5714

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
           0      0.71429       1.4286       2.1429       2.8571       3.5714       4.2857
          10       11.429       12.857       14.286       15.714       17.143       18.571
```

```
 Column 15:

          20
           5
          20


//////////SUM/////////

A1sum =   750
A2sum1 =

 Columns 1 through 7:

          -4        -0.5           3         6.5          10        13.5          17

 Columns 8 through 14:

        20.5          24        27.5          31        34.5          38        41.5

 Column 15:

          45

A2sum2 =

 Columns 1 through 7:

          -4        -0.5           3         6.5          10        13.5          17

 Columns 8 through 14:

        20.5          24        27.5          31        34.5          38        41.5

 Column 15:

          45

A2sum3 =

       157.5
   3.5527e-15
         150


//////////PROD/////////

A1prd = 4.7131e+40
A2prd1 =

 Columns 1 through 7:

           0     -14.431     -37.901     -62.099     -78.717     -79.446     -55.977
```

```
 Columns 8 through 14:

           0       96.793       242.71       446.06       715.16       1058.3       1483.8

 Column 15:

         2000

A2prd2 =

 Columns 1 through 7:

           0      -14.431      -37.901      -62.099      -78.717      -79.446      -55.977

 Columns 8 through 14:

           0       96.793       242.71       446.06       715.16       1058.3       1483.8

 Column 15:

         2000

A2prd3 =

   5.0004e+13
           0
           0


/////////CUMSUM/////////

A1csum =

 Columns 1 through 7:

          -30      -56.207      -78.621      -97.241      -112.07       -123.1      -130.34

 Columns 8 through 14:

      -133.79      -133.45      -129.31      -121.38      -109.66      -94.138      -74.828

 Columns 15 through 21:

      -51.724      -24.828       5.8621       40.345       78.621       120.69       166.55

 Columns 22 through 28:

       216.21       269.66        326.9       387.93       452.76       521.38       593.79

 Columns 29 and 30:

          670          750
```

```
A2csum1 =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -4      -1.9286      0.14286       2.2143       4.2857       6.3571       8.4286
          -4         -0.5            3          6.5           10         13.5           17

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
        10.5       12.571       14.643       16.714       18.786       20.857       22.929
        20.5           24         27.5           31         34.5           38         41.5

 Column 15:

          20
          25
          45

A2csum2 =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -4      -1.9286      0.14286       2.2143       4.2857       6.3571       8.4286
          -4         -0.5            3          6.5           10         13.5           17

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
        10.5       12.571       14.643       16.714       18.786       20.857       22.929
        20.5           24         27.5           31         34.5           38         41.5

 Column 15:

          20
          25
          45

A2csum3 =

 Columns 1 through 7:

           1       3.3571       7.0714       12.143       18.571       26.357         35.5
          -5      -9.2857      -12.857      -15.714      -17.857      -19.286          -20
           0       1.4286       4.2857       8.5714       14.286       21.429           30

 Columns 8 through 14:

          46       57.857       71.071       85.643       101.57       118.86        137.5
         -20      -19.286      -17.857      -15.714      -12.857      -9.2857           -5
          40       51.429       64.286       78.571       94.286       111.43          130
```

```
 Column 15:

       157.5
   3.5527e-15
         150


////////CUMPROD/////////

A1cprd =

 Columns 1 through 7:

         -30       786.21       -17622   3.2813e+05  -4.8654e+06   5.3687e+07  -3.8877e+08

 Columns 8 through 14:

   1.3406e+09   4.6227e+08   1.9128e+09   1.5171e+10   1.7786e+11      2.76e+12   5.3296e+13

 Columns 15 through 21:

   1.2313e+15   3.3118e+16   1.0164e+18   3.5048e+19   1.3415e+21   5.6435e+22   2.5882e+24

 Columns 22 through 28:

   1.2852e+26   6.8691e+27    3.932e+29   2.3999e+31   1.5558e+33   1.0676e+35   7.7308e+36

 Columns 29 and 30:

   5.8914e+38   4.7131e+40

A2cprd1 =

 Columns 1 through 7:

           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -5      -10.102      -13.265       -14.49      -13.776      -11.122      -6.5306
           0      -14.431      -37.901      -62.099      -78.717      -79.446      -55.977

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
           0       8.4694       18.878       31.224        45.51       61.735       79.898
           0       96.793       242.71       446.06       715.16       1058.3       1483.8

 Column 15:

          20
         100
        2000

A2cprd2 =

 Columns 1 through 7:
```

```
           1       2.3571       3.7143       5.0714       6.4286       7.7857       9.1429
          -5      -10.102      -13.265       -14.49      -13.776      -11.122      -6.5306
           0      -14.431      -37.901      -62.099      -78.717      -79.446      -55.977

 Columns 8 through 14:

        10.5       11.857       13.214       14.571       15.929       17.286       18.643
           0       8.4694       18.878       31.224        45.51       61.735       79.898
           0       96.793       242.71       446.06       715.16       1058.3       1483.8

 Column 15:

          20
         100
        2000

A2cprd3 =

 Columns 1 through 7:

           1       2.3571       8.7551       44.401       285.43       2222.3        20318
          -5       21.429      -76.531       218.66      -468.55       669.36      -478.12
           0            0            0            0            0            0            0

 Columns 8 through 14:

   2.1334e+05   2.5296e+06   3.3427e+07   4.8708e+08   7.7585e+09   1.3411e+11   2.5002e+12
           0            0            0            0            0            0            0
           0            0            0            0            0            0            0

 Column 15:

   5.0004e+13
           0
           0


//////////DIFF//////////

A1df =

 Columns 1 through 7:

       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931

 Columns 8 through 14:

       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931

 Columns 15 through 21:

       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931
```

```
 Columns 22 through 28:

      3.7931       3.7931       3.7931       3.7931       3.7931       3.7931       3.7931

 Column 29:

      3.7931

A2df =

 Columns 1 through 7:

          -6      -6.6429      -7.2857      -7.9286      -8.5714      -9.2143      -9.8571
           5       5.7143       6.4286       7.1429       7.8571       8.5714       9.2857

 Columns 8 through 14:

       -10.5      -11.143      -11.786      -12.429      -13.071      -13.714      -14.357
          10       10.714       11.429       12.143       12.857       13.571       14.286

 Column 15:

         -15
          15

Bazhaete prodovzhiti?(Y/N) n
Prohrama zavershena.
```

4 Висновок

Під час виконання лабораторної роботи я навчився використовувати методи статистичної обробки даних в середовищі GNU Octave. Від самого початку постала проблема виведення результатів обрахунку, адже результат був настільки великим, що користувач навряд зміг би там щось знайти. Дану проблему я вирішив оператором “Switch”, та циклом “While”. “Switch” дозволяє нам зручно обирати з якої “Теми” нам потрібні розрахунки, а “While” дає можливість одразу перезапустити програму та обрати інший “Case”. На початку кожного “Case” стоїть команда “clc” що очищує командне вікно від попередніх розрахунків (результати обрахунків не стираються а записуються в змінні у Workspace). Також відкриттям для мене стала функція “strcmpi()”, що порівнює значення двох рядків без врахування регістру символів (великих і малих літер).